

Nonlinear Parameter Recovery from Modal Feature Frequencies

Inversion infrastructure, Jacobian-predicted feature tracking, a noise-free multistart twin inversion, and feature-perturbation sensitivity

Technical writeup

This note documents the EMM inversion stage: the code built to invert the five modal feature frequencies for (β, γ, R) , the change to feature tracking that the first experiments forced, a noise-free twin inversion from several starts, and a controlled study of how feature errors propagate into parameter errors. It follows the Jacobian report, whose results are summarised in Section 1. Colour convention: findings in green, open or scope-limited points in red, definitions and methods in blue.

1 Starting point

Features, reference point and prior local result. The forward map is $f(\beta, \gamma, R) \in \mathbb{R}^5$, the sub-grid locations of five features of the EMM response projected onto dry plate modes, in the fixed order $[A_{2,0}, A_{0,1}, A_{1,1}, A_{2,1}, A_{0,0}]$: three maxima and two near-zeros. Each is the vertex of a quadratic fitted to $|A_{n,j}|^2$ at three anchor frequencies, with strict validity checks (degenerate, extrapolated, kindMismatch). At the reference (test) parameter set and its feature vector are

$$\begin{aligned} p_0 &= [\beta_0, \gamma_0, R_0] = [4.6985 \times 10^{-5}, 1.4548 \times 10^{-3}, 0.3830], \\ f_0 &= [5.836, 5.677, 6.648, 4.959, 7.846] \text{ rad/s}, \end{aligned} \quad (1)$$

with anchors (rad/s) $A_{2,0}$: [5.70, 5.80, 5.90]; $A_{0,1}$: [5.50, 5.65, 5.80]; $A_{1,1}$: [6.45, 6.60, 6.75]; $A_{2,1}$: [4.90, 4.95, 5.00]; $A_{0,0}$: [7.80, 7.85, 7.90]. The scaled Jacobian $J_s = \partial \ln f / \partial \ln p$ is full rank with condition number $\kappa \approx 293$ and a weakest direction that is almost pure γ . p_0 is a test point for method development, not a claim about a real floe. Parameters are handled in scaled form $x = p/p_0$ throughout.

2 Code built in this stage

Every function was tested against analytic toy problems before touching the EMM, and all code in **Parameter Inversion/** is forward-model-agnostic: the EMM enters only through a function handle, so the same machinery will accept an SSI-derived feature map later.

File	Role	Outcome
<code>inverseResidual.m</code>	Relative residual $r = (f(x \odot p_0) - f_{\text{obs}}) / f_{\text{obs}}$; optional feature mask.	Tester: 6 groups, 25 invalid-input cases, all pass.
<code>solveInverse.m</code>	Bounded <code>lsqnonlin</code> wrapper: central differences at step 10^{-4} , tight tolerances, iterate logging, expected-error classification.	Tester: 6 groups pass; step 10^{-4} confirmed from recorded evaluation points.
<code>probeValidRegion.m</code>	Maps where the strict forward map succeeds: axis probes, corner checks, conservative bounds.	Tester: 8 groups pass; bounds match hand derivation.
<code>predictFeatureAnchors.m</code>	Moves each feature's anchors to its Jacobian-predicted location.	Tester: 4 groups pass, incl. real reference point.
<code>modalFeatureVector.m</code> (modified)	Option <code>AnchorMode</code> : fixed (default, unchanged) or predicted .	Existing tester still passes; fixed mode reproduces f_0 .
<code>runProbeValidRegionEMM.m</code>	Probe driver, either anchor mode.	Sections 3, 4.
<code>runEMMTwinInversion.m</code>	Noise-free multistart twin.	Section 5.
<code>runFeaturePerturbationStudy.m</code>	Linear noise propagation, worst-direction test, nonlinear verification.	Section 6.

Table 1: Files added or changed in this stage and their outcomes.

Two solver details that matter. (i) `lsqnonlin`'s default finite-difference step is about 1.5×10^{-8} , three orders of magnitude below where the feature derivatives were validated; `solveInverse` forces

central differences at 10^{-4} in scaled units, the validated relative step. (ii) An error thrown by the strict forward map aborts `lsqnonlin`. `solveInverse` records errors whose identifiers the caller declares as “left the valid region” and re-throws everything else, so a bug is never recorded as a finding.

3 Fixed anchors fail as a nonlinear feature tracker

The fixed anchors cannot tolerate a 1% change in R . Probing each parameter at $\pm 1, 2, 5, 10$ and 20% around p_0 with the fixed anchors gave the ranges in Table 2. Every failure was an **extrapolated** fit, not a solver failure: the feature moved outside its frozen bracket.

Parameter	Probe: verified range	Linear prediction from J_s	Limiting feature
β	-5% to $+10\%$ (fails $-10\%, +20\%$)	-8.0% to $+9.3\%$	$A_{0,0}$
γ	$\pm 20\%$ (no failure)	about $\pm 85\%$	$A_{0,0}$
R	fails at $\pm 1\%$	-0.86% to $+0.74\%$	$A_{0,0}$

Table 2: Fixed-anchor probe versus the first-order prediction from J_s , f_0 and the anchor brackets.

The linear prediction reproduces the probe without further EMM calls: R is the strongest parameter direction, so features move fastest in R and leave narrow brackets first. At box corners the same model allows only about $\beta \pm 2\%$, $\gamma \pm 10\%$, $R \pm 0.3\%$. The fixed anchors are a good local derivative extractor but not a usable tracker for a nonlinear inversion.

4 Jacobian-predicted anchors

Predicted anchors. At a trial p , each feature’s three anchors are translated rigidly (spacing unchanged) by the log-linear prediction

$$f_{\text{pred}} = f_0 \odot \exp(J_s \ln(p/p_0)), \quad \omega_{\text{anchors}}^{(k)}(p) = \omega_{\text{anchors}}^{(k)}(p_0) + (f_{\text{pred},k} - f_{0,k}). \quad (2)$$

The prediction only decides where to look: the returned feature is still the vertex of the quadratic fitted to the EMM field at the moved anchors, with all three validity checks judged against the moved bracket. The map is smooth in p , costs the same 15 EMM solves, and at $p = p_0$ reproduces fixed mode exactly. p_0 , f_0 and J_s (at $\epsilon = 10^{-4}$) are saved once in `anchorPredictionModel.mat`.

Tested, then validated on the real EMM. Predicted mode equals fixed mode bit for bit at p_0 ; finite differences through predicted mode show no jumps ($|D(h) - D(2h)|/|D| \leq 6.1 \times 10^{-5}$); and the R -column of the Jacobian measured through predicted anchors agrees with the fixed-anchor J_s to within 0.3–3.1% (Table 3), the small difference expected between two estimators of one derivative.

	$A_{2,0}$	$A_{0,1}$	$A_{1,1}$	$A_{2,1}$	$A_{0,0}$
J_s , fixed anchors	−0.4184	−0.2644	−0.2580	−0.6927	−0.7965
Measured, predicted anchors	−0.4171	−0.2670	−0.2661	−0.6741	−0.7832
Difference	0.31%	0.99%	3.1%	2.7%	1.7%

Table 3: R -column of the scaled Jacobian at p_0 , fixed versus predicted anchors.

The valid region expands by more than an order of magnitude in R . With predicted anchors (39 evaluations, 24.4 min): β and γ valid at every tested point to $\pm 20\%$, R valid to $\pm 10\%$ (first failure at $\pm 20\%$, $A_{2,1}$ extrapolated), and all eight corners of

$$x \in [0.84, 0.84, 0.92] \dots [1.16, 1.16, 1.08] \quad (3)$$

valid. Along the axes the minimum bracket margin stayed near 0.32 (its value at p_0), falling only to 0.18 at $R = \pm 10\%$: the prediction keeps each feature at almost the same position inside its bracket.

Admissible, but tight at one corner. At $x = [1.16, 0.84, 0.92]$ the $A_{0,0}$ margin is 0.031, so Equation (3) is treated as the outer demonstrated region and the inversion uses a smaller box inside it. The γ axis is independent evidence of weak γ sensitivity, since the margin is measured from the real EMM vertex: over $\pm 20\%$ in γ it moved by about 0.002. The limitation is locality and one-dimensionality, not circularity.

5 Noise-free multistart twin inversion

Setup. Hidden truth $x_{\text{true}} = [1.08, 0.93, 1.04]$, all three parameters perturbed; scaled bounds 0.90–1.10 (β), 0.85–1.15 (γ) and 0.95–1.05 (R), inside Equation (3), with the truth and every start at least 5% of the box width from every bound. $f_{\text{obs}} = f(p_{\text{true}})$ is generated once with predicted anchors and the same map is inverted: a deliberate same-model (“inverse crime”) integration test. Bracket margins at the truth, by feature: 0.319, 0.416, 0.348, 0.371 and 0.495.

Start x_0	Rel. error β	Rel. error γ	Rel. error R	Resnorm	Iter. / calls / min
[1.00, 1.00, 1.00]	$+5.2 \times 10^{-7}$	$+1.7 \times 10^{-5}$	-4.1×10^{-8}	3.8×10^{-15}	8 / 63 / 46
[0.95, 1.08, 1.03]	-4.1×10^{-8}	-9.9×10^{-7}	$+2.2 \times 10^{-9}$	1.5×10^{-17}	7 / 56 / 37
[1.05, 0.90, 0.97]	$+2.0 \times 10^{-7}$	$+6.6 \times 10^{-6}$	-1.6×10^{-8}	5.8×10^{-16}	9 / 70 / 104
Spread across starts	6.1×10^{-7}	1.65×10^{-5}	4.5×10^{-8}		

Table 4: Twin recovery. All three starts ended with status `solved` (first-order optimality). A fourth planned start was not run once three had agreed.

All three parameters recovered, from three different starts, to the same point, and the weak direction is visible even in exact data. β and R approach the truth almost directly, while γ follows non-monotonic, overshooting trajectories (Table 5): the optimiser reduces the residual by two to four orders of magnitude before γ turns around. The final γ error is roughly 25–450 times the β and R errors and varies between starts, set by where the stopping tolerance acts: a final feature residual of about 3×10^{-8} divided by σ_3 gives about 10^{-5} in γ , as observed. The scaled Jacobian at the solution has $\sigma = [1.161, 0.1108, 0.003677]$, $\kappa \approx 316$ and weak vector $[-0.031, -0.9995, +0.002]$, essentially the reference-point result.

Start	γ/γ_0 by iteration (truth 0.930)
1	1.000 $\rightarrow$ 1.048 $\rightarrow$ 1.083 $\rightarrow$ 1.094 $\rightarrow$ 1.067 $\rightarrow$ 0.975 $\rightarrow$ 0.941 $\rightarrow$ 0.931 $\rightarrow$ 0.930
2	1.080 $\rightarrow$ 0.957 $\rightarrow$ 0.905 $\rightarrow$ 0.887 $\rightarrow$ 0.886 $\rightarrow$ 0.918 $\rightarrow$ 0.930 $\rightarrow$ 0.930
3	0.900 $\rightarrow$ 0.986 $\rightarrow$ 1.038 $\rightarrow$ 1.076 $\rightarrow$ 1.086 $\rightarrow$ 1.048 $\rightarrow$ 0.967 $\rightarrow$ 0.938 $\rightarrow$ 0.931 $\rightarrow$ 0.930

Table 5: γ iterates; turning points in bold. The overshoot direction depends on the start.

6 Feature-perturbation sensitivity

Method. From the solved twin, $J_{\text{rel}} = \partial r / \partial \ln x$ at the solution (from `lsqnonlin`’s Jacobian, no EMM calls). Baseline noise model: $f_{\text{obs}} = f_{\text{true}} \odot (1 + \eta)$, $\eta_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_f^2)$, the same relative σ_f on all five features; the residual at the truth is $d = -\eta/(1 + \eta)$ and $\delta \ln x = -J_{\text{rel}}^+ d$. 10^4 samples at each of seven levels, checked against the first-order linear-Gaussian benchmark $\sigma_f \sqrt{\text{diag}(J_{\text{rel}}^+ J_{\text{rel}}^{+T})}$ (agreement within 1.7%). This is a controlled sensitivity study, not a model of real SSI errors.

Noise amplification per parameter. $\text{RMS}(\delta \ln x)/\sigma_f = [12.35, 271.8, 1.12]$ for (β, γ, R) : γ is about 22 times more sensitive than β and 243 times more sensitive than R .

σ_f	$\text{RMS } \delta \ln \beta$	$\text{RMS } \delta \ln \gamma$	$\text{RMS } \delta \ln R$	γ predicted outside box	any
10^{-5}	1.2×10^{-4}	2.7×10^{-3}	1.1×10^{-5}	0.0%	0.0%
10^{-4}	1.2×10^{-3}	2.7×10^{-2}	1.1×10^{-4}	0.1%	0.1%
3×10^{-4}	3.7×10^{-3}	8.2×10^{-2}	3.3×10^{-4}	13.7%	13.7%
10^{-3}	1.3×10^{-2}	0.27	1.1×10^{-3}	57.9%	59.8%
10^{-2}	0.12	2.7	1.1×10^{-2}	95.4%	97.7%

Table 6: Linear Monte Carlo (selected levels of the seven run). The “outside box” columns are linearised prediction rates under the assumed noise model, not nonlinear inversion failure probabilities. Rows at 10^{-3} and above indicate breakdown of the local inverse rather than quantitative predictions.

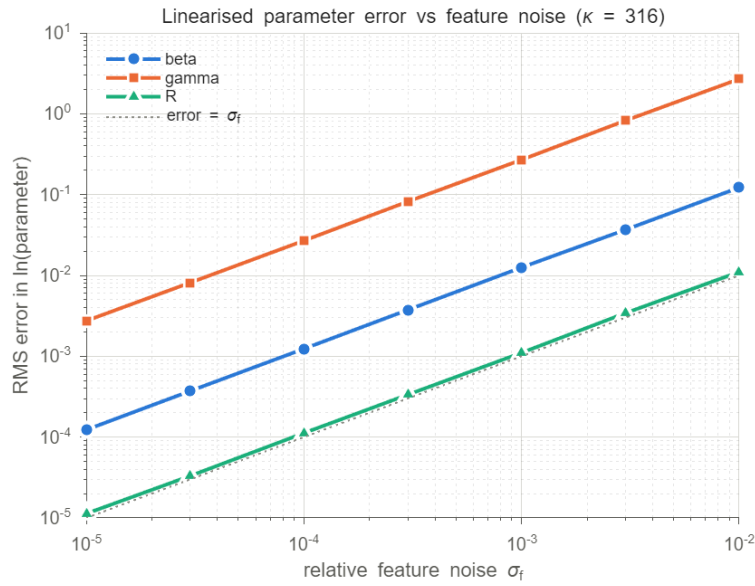


Figure 1: Linearised RMS parameter error versus relative feature noise. R tracks σ_f almost one to one; γ sits more than two decades higher.

Worst direction, and the nonlinear inverse follows it. A residual $s a u_3$ at the truth (u_3 the weakest left singular vector) gives, to first order, $\delta \ln x = -s(a/\sigma_3)v_3$. Full nonlinear inversions on $f_{\text{obs}} = f_{\text{true}}/(1 + s a u_3)$, warm-started at the truth, agree with that prediction to within 4% up to $a = 3 \times 10^{-4}$ (Table 7). The departure is systematic: $+u_3$ moves γ slightly less than predicted and $-u_3$ slightly more, with the gap growing roughly in proportion to a (about 2.6% at 10^{-4} , 6.7% at 3×10^{-4}), the signature of a second-order term. At $a = 3 \times 10^{-4}$, sign -1 , the nonlinear inverse follows the prediction to $\gamma/\gamma_0 = 0.855$, just above the lower bound of 0.85. The $a = 10^{-3}$ pair was not run: its linear prediction ($\gamma/\gamma_0 \approx 1.220$ or 0.709) lies outside the box on both sides.

a	sign	Linear γ/γ_0	Nonlinear γ/γ_0	$\delta \ln \gamma$ ratio NL/lin	Iter. / calls	Status
10^{-4}	+1	0.9556	0.9553	0.987	3 / 28	solved, interior
10^{-4}	-1	0.9051	0.9047	1.013	3 / 28	solved, interior
3×10^{-4}	+1	1.0090	1.0059	0.962	3 / 28	solved, interior
3×10^{-4}	-1	0.8572	0.8552	1.029	5 / 42	solved, interior

Table 7: Nonlinear verification along the weakest feature direction (β and R agreed with their linear predictions to within 3×10^{-5} in scaled units in every case).

7 Which features carry the information

Dropping rows of J_{rel} shows which features constrain which parameter (Table 8). Every subset of three or more features keeps numerical rank three, so the question is practical conditioning, not rank. The condition number alone is a poor ranking metric: removing $A_{2,0}$ lowers κ from 316 to 301 yet slightly worsens every parameter, because a row that mainly strengthens the strong direction raises σ_1 and hence κ while still reducing variance. $\sigma_{\min}$ and the per-parameter factors are the meaningful measures.

Features	$\sigma_{\min}$	κ	RMS/ σ_f (β, γ, R)	Note
all five	0.00368	316	12.4, 272, 1.12	reference
$A_{0,1}, A_{1,1}, A_{2,1}, A_{0,0}$	0.00360	301	12.4, 278, 1.27	drop $A_{2,0}$
$A_{2,0}, A_{0,1}, A_{2,1}, A_{0,0}$	0.00367	308	14.2, 272, 1.22	drop $A_{1,1}$
$A_{2,0}, A_{1,1}, A_{2,1}, A_{0,0}$	0.00335	337	16.6, 299, 1.14	drop $A_{0,1}$
$A_{2,0}, A_{0,1}, A_{1,1}, A_{0,0}$	0.00174	549	23.7, 576, 2.66	drop $A_{2,1}$
$A_{2,0}, A_{0,1}, A_{1,1}, A_{2,1}$	0.00104	837	14.5, 960, 1.17	drop $A_{0,0}$
$A_{0,1}, A_{2,1}, A_{0,0}$	0.00356	295	14.9, 280, 1.50	best three
$A_{1,1}, A_{2,1}, A_{0,0}$	0.00334	313	17.3, 299, 1.33	
$A_{2,0}, A_{2,1}, A_{0,0}$	0.00310	355	28.2, 322, 1.49	
$A_{2,0}, A_{0,1}, A_{1,1}$	0.00103	553	23.8, 967, 3.08	maxima only
$A_{2,0}, A_{0,1}, A_{2,1}$	0.00037	2242	14.5, 2691, 1.28	worst three

Table 8: Feature-subset conditioning from the full-precision J_{rel} at the solution (all five four-feature subsets and selected three-feature subsets of the ten computed).

The two near-zero features hold the γ information, jointly. The maxima-only set keeps rank three but raises the γ amplification from $272\sigma_f$ to $967\sigma_f$. Adding $A_{2,1}$ alone barely helps (960), adding $A_{0,0}$ alone roughly halves it (576), and only the pair restores ≈ 272 , consistent with the weakest feature direction $u_3 \approx [-0.19, 0.32, -0.03, -0.70, 0.60]$, dominated by $A_{2,1}$ and $A_{0,0}$ with opposite signs. Every three-feature subset containing both near-zeros stays within $280\text{--}322\sigma_f$. $A_{2,1}$ also carries information on R (dropping it raises the R factor from 1.12 to 2.66), and $A_{2,0}$ is nearly expendable.

Consequence for the SSI bridge. This is a subset analysis of the EMM feature Jacobian: it does not show that SSI can recover any of these features, nor that an SSI-derived feature vector will have the same sensitivities. It does show what the bridge must replace. If SSI supplies only pole frequencies analogous to the three maxima, the three-parameter inverse stays formally identifiable but γ becomes very fragile. Candidate SSI observables that might carry the missing information include damping ratios (added mass and radiation damping depend on the mass/draft parameter) and mode shapes or relative amplitudes at the sensor positions.

8 Discussion

Jacobian-predicted feature tracking is valid over $\beta, \gamma \pm 16\%$, $R \pm 8\%$. Within it the nonlinear inverse recovers all three parameters from exact data, from several starts, to the same point. Under feature error, β and R stay well determined while γ amplifies relative feature error by about 270, a figure the nonlinear inverse confirms to within 4% along the worst direction.

Newly established: the generic, tested inversion code; the failure mechanism of the fixed anchors, explained quantitatively by J_s ; a smooth replacement widening the usable region by more than an order of magnitude in R ; noise-free recovery from three starts; the weak γ direction persisting at the perturbed truth ($\kappa \approx 316$); a quantitative noise-to-parameter map verified nonlinearly; and the dependence of γ on the two near-zero features.

Not yet established. (1) Everything is local to one test point and one truth. (2) The noise model is idealised (iid, equal relative error on every feature). (3) All observables are EMM-side forced-response features; SSI returns poles and modes of the wet, damped system, which are not automatically the same quantities.

Next steps. First, the SSI bridge: define an SSI-compatible feature vector $f_{\text{SSI}}(\beta, \gamma, R)$ computable from the forward model (frequencies, and candidates such as damping and mode-shape measures), build its Jacobian, and compare it against Table 8, with the recoverability of $A_{0,0}$ -like information as the first question. Second, raise with the supervisors whether γ should be supplied from measured mass and planform area, $\gamma = (m/A)/(\rho_w H)$, leaving (β, R) to the frequency inversion. The results above make that option quantitatively well motivated; the baseline twin deliberately kept all three unknowns.